Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Fisher, John W M.S.		
Project Title:	SBIR Phase II Functionality & Operations Testing of AFR (Advanced Fuel Research) Dry Pyrolysis and Enertech Wet Pyrolysis Instruments		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Moffet Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Pyrolysis processing is one of several options for solid waste resource recovery in space. It has the advantage of being relatively simple and adaptable to a wide variety of feedstocks and it can produce several usable products from typical waste streams. This task agreement provides for advancement of pyrolysis technology for waste processing in space. In FY04 the hardware obtained from a phase II SBIR will be tested and research will be conducted to reduce the tars produced by pyrolysis os space mission type wastes. A wet pyrolysis system made by Enertech Environmental Inc. has been delivered to NASA Ames. This hardware will be installed and tested to evaluate its ability to convert carbonaceous waste in high solids content slurries that can be transferred in a waste management system by pumping. A series of test on a dry pryrolysis system made by Advanced Fuels Research Inc. has revealed problems associated with		

	tar formation. Further tests and development will be conducted in order to reduce or eliminate the tar problem.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth benefits include applications for conversion of waste materials to fuels, safe disposal of hazardous biological wastes, and more efficient disposal of general waste materials.
Task Progress:	No progress report received this reporting period. Place progress report here.
Bibliography Type:	Description: (Last Updated: 02/27/2004)